

Smart Traffic Management at Busy Traffic Light Junctions across India using Internet of Things and Artificial Intelligence

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Abstract - This paper deals with the idea of using the concepts like internet of things and artificial intelligence in traffic signal control for better and more efficient traffic management at busy traffic light junctions in all big cities across india. The paper tries to demonstrate the audience how use of Artificial intelligence and internet of things in traffic lights at junctions can be used to prevent inefficient flow of traffic through a junction. We can record the flow and density of traffic coming from each road constituting the junction at different times of the day (i.e. regular hours and peak hours) and can develop a system that will analyze this data in real time. All The traffic signals at a junction can then be connected to this system which will link them together forming a network of traffic lights sharing data, which will enable them to work in sync with each other. The sequence of operation and duration of each traffic light at the junction will be controlled by the system to ensure a consistent flow of traffic in all directions through the junction depending upon the real time traffic density. With the help of Artificial intelligence the system can learn and modify the timing and operation sequence to maximize efficiency. The use of this system will help in solving problems like slow moving traffic, unnecessary road congestion, vehicular pollution, delays and increased travel time. It will help improve the existing traffic conditions and enhance the overall travel experience.

Keywords - Artificial intelligence, Internet of things, Traffic congestion, Traffic signal control.

I. Introduction

With the rise in population, the demand for transportation is also increasing rapidly every year. Due to the increasing

number of vehicles and overuse of roads traffic jam is becoming a major issue in all the major cities across India.

Traffic Jam has a tremendous impact on the life of people. Vehicles have to wait for a long time to move in and out of the jam. It is one of the most serious problems in big cities that people have to deal with in daily life. Often it is caused due to narrow roads, poor traffic management and increased number of vehicles. India has the second largest road network in the world covering about 5.4 million kilometers. Major cities like Delhi, Bangalore, Mumbai and many other cities of India are facing the issue of traffic congestion. The standard speed of many vehicles on the road of Indian cities is less than 20 kilometers per hour, and a journey of 10 kilometers could take 30 minutes or more. These traffic congestions cause a huge wastage of time and energy. The big cities of India are losing up to \$22 billion every year to traffic congestion, and this burden is carried by the commuters of these cities.



Fig 1: Traffic congestion at a busy Junction

India's road network has grown at an annual rate of 4% since 1951 but the number of vehicles has increased nearly 11%, choking roads and increasing pollution. In order to make traffic flow as efficient as possible and to make the most of our present road network we need to upgrade our traffic control system especially at traffic junctions. With the use of emerging technologies like 'Internet of things' and 'artificial intelligence' our traffic systems can be improved to cope with the increasing number of vehicles and to manage traffic better and more efficiently.

1.1. Internet of things

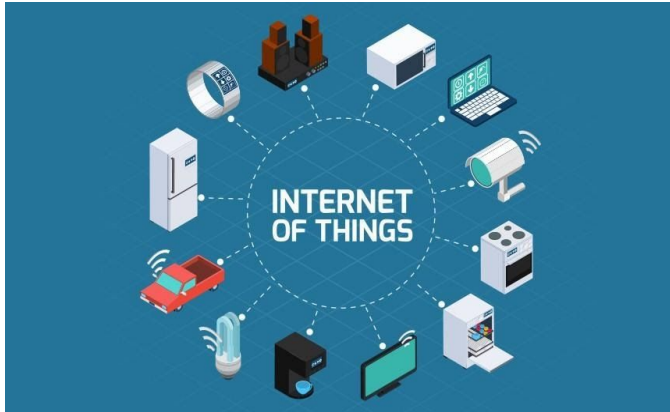


Fig 3: Internet of Things

The Internet of Things is the network of physical objects like devices, vehicles, buildings and other items—embedded with electronics, softwares, sensors, and network connectivity that enables these objects to collect and exchange data. In simple words, the IoT allows objects to be sensed and controlled remotely across existing network infrastructure, creating opportunities for more direct integration of the physical world into computer-based systems.

1.2. Artificial intelligence

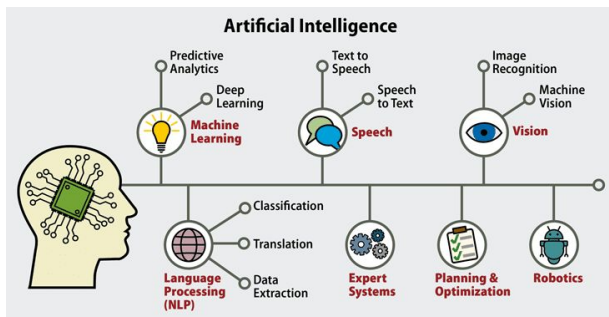


Fig 2: Diagram Explaining Artificial Intelligence

Artificial intelligence is the phenomenon which refers to the ability of a machine to interpret and learn from external data. The machine can make different decisions based on this

data on its own and can learn to recognize a pattern that it perceives in its environment and takes actions that maximize its chance of successfully achieving its goal. Artificial intelligence can be used to recognize objects with the help of cameras and image recognition software.

II. Existing Methods

Many different ideas and methods have been proposed for monitoring and controlling traffic over the period of time.

In [1] the survey paper presents the various urban traffic control strategies which have been

proposed or applied to prevent and treat congestion. Some basic definitions about congestion, such as primary and secondary congestion, are remembered. Freeway control during congested situations is examined; the access control and exits control problems are developed. Approaches for intersection control are described: delay minimization, output flow rates maximization, LQ problem, congestion level feedback and democratic control.

Going forward towards a novel approach, it has been proposed to use measurements from a single station to detect anomalies in traffic flow. It was found that Anomalies are ascertained from deviations from the expected norms of traffic patterns calibrated at each individual station. This [2] paper discusses the development of a new novel time indexed anomaly detection algorithm. The author established norms as a time dependent function for each station by integrating the past "normal" traffic patterns for a given time period.

In the changing times keeping into consideration the existing traffic, a viable solution has been laid into place which will not only help the tracker to take the active control of the congestion but at the same time doing it in the most efficient manner. This paper [3] describes implementation of a real-time Visual tracking system equipped with an active camera.

The system is intended for indoor human motion tracking.

Real-time tracking is achieved using simple and fast motion detection procedures based on frame differencing and camera motion compensation. Results of on-line person tracking are presented.

In the contemporary times a more mechanized method has been introduced to channelize the control of traffic. In this paper [4] Background subtraction is a common computer vision task. The author analyzes the usual pixel-level approach. The author also develops an efficient adaptive algorithm using Gaussian mixture probability density. Recursive equations are used to constantly update the parameters but also to simultaneously select the appropriate number of components for each pixel.

In [5] a logical alternative was set to be proposed at the outset which would instil a new system for wireless traffic light controller, it makes a traffic policeman easily and effectively control a road junction with a wireless remote control. The proposed system helps the policeman to control the junction by himself and be able to change conditions of the traffic flow dynamically.

However, over a significant time period it was into consideration present an intelligent Traffic Congestion Monitoring & Measurement System called Traffic Monitor to monitor and measure the road traffic congestions using probe vehicle. The concept of probe vehicle has come up in recent times for collecting real time traffic data. The system [6] provides an easy platform to analyze the traffic movement and congestion pattern.

In this paper [7] the author proposes an accurate and fast foreground detection technique for object tracking in videos with quasi-stationary backgrounds. The background is modeled using a novel real-time kernel density estimations approach based on online histogram learning.

In the times of dynamism a possible alternative has come into place which proposes to suggest an intelligent traffic system. In this paper [8] a model is also proposed capable of managing intelligent traffic system using CCTV cameras and WAN. The proposed model will make the traffic signaling dynamic and automatic as well. Besides this, it will generate the dynamic messages for the users on the message boards to avoid congestion, reduce waiting time, pollution control, accident control and vehicle tracking.

Paving the way for further well planned method the author [10] proposed the algorithm to determine the number of vehicles on the road. The density counting algorithm works by comparing the real time frame of live video by the reference image and by searching vehicles only in the region of interest (i.e. road area). The computed vehicle density can be compared with other direction of the traffic in order to control the traffic signal smartly.

Further computation was taken into place and the author [11] devised a method based on optical flow calculation and Delaunay triangulation. Given a sequence of frames, firstly, the author extracted the corner feature points using ORB algorithm and compute optical flow vectors at the extracted feature points. Secondly, the author separates the optical flow vectors using K-Means clustering method. Third, we classify each cluster into camera and object motion using its mean scatter value. Finally, we represent the moving object using Delaunay triangulation.

The idea of presenting a well planned application for traffic management has been suggested by the author [12] when he investigates the prototype development of Internet of Things (IoT) application for traffic management.

The retrofitted traffic light control solution is proposed and developed to support decision making of police officers. The system can figure out the congestion level of each road at a specific intersection with the help of RFID technology.

To set the basics right, and devise a method which has a better tenacity the author has proposed this research. The paper [16] presents novel intelligent traffic administration system, based on Internet of Things, which is featured by low cost, high scalability, high compatibility, easy to upgrade, to replace traditional traffic management system and the proposed system can improve road traffic tremendously.

To cater to the emerging and urgent need of the mushrooming population about the traffic systems, the author has proposed this paper. The paper [18] analyses the ever growing urban population around the globe and discusses about the traffic systems in densely populated cities like Los Angeles and Amsterdam. Further, an advanced traffic management system is proposed, implemented using Internet of Things (IoT). The system is supported by a circuit embedded in the vehicle, which operates using RFID with clustered systems.

III. Proposed idea for developing an AI & IOT based traffic lights system to manage traffic flow at busy junctions in major cities of India

Traffic congestion at junctions is caused due to many factors such as improper junction design, narrow roads, heavy traffic, etc. but all these factors are circumstantial and are not present at every junction. The one factor that is present at every junction with a congestion problem is poor traffic management due to inefficient traffic signal control. This problem arises because the current traffic signal lights are preprogrammed and works on a fixed timer. The traffic signals at junctions are also not linked together and operate independently. They don't account for the fluctuations in traffic density taking place on all the roads in real time. For example, the light turns red for 30 seconds and then green for 30 seconds irrespective of the number of vehicles present at the signal and without taking into consideration the traffic flow on other roads connected to the junction and how it will affect it. Due to this congestions and delays take place at the junction.

In order to prevent traffic congestions at junctions and to make sure a smooth and efficient flow of vehicles through a junction, each road connected to the junction should have a traffic light at least half a kilometer before the junction's main traffic lights as shown in **Fig 4**, let's call these lights 'secondary lights' and the traffic lights present at the junction

‘primary lights’ for easier understanding. Secondary lights will be used to control the number of cars approaching the junction from every road at a given time just like a control valve and the primary lights will allow the cars to move through the junction. This way all the traffic on the roads will not pile up on the junction at once thereby causing congestion.

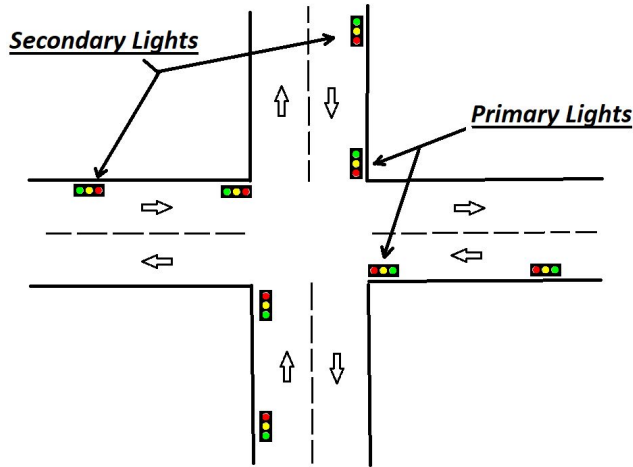


Fig 4: Concept of Primary & Secondary Lights

All the primary and secondary traffic lights will be connected to a central server forming a network. Cameras placed near the primary and secondary lights can be used to keep a track of number of vehicles present using real time image recognition. This data will be uploaded to the central server controlling the junction traffic light network. Based on the traffic movement and density at the primary lights, the secondary lights will allow traffic movement towards the junction. The primary lights turn green allowing the traffic to move through the junction. Once the traffic clears the secondary lights will allow a controlled number of vehicles to move towards the junction into the controlled traffic area as Shown in Fig 5.

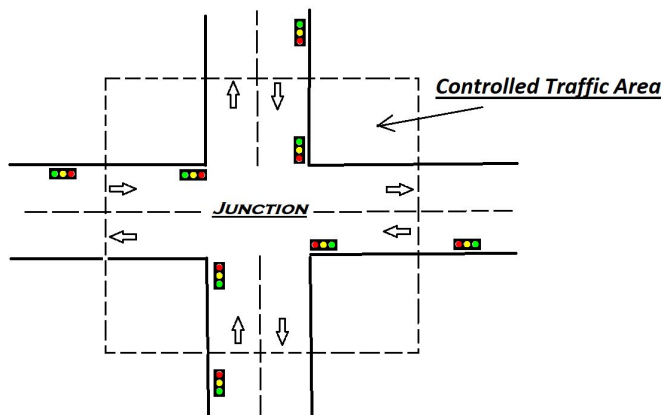


Fig 5: Traffic Controlled Area

This technique will be applied on all the roads connected to a junction. This way only that many vehicles will be present at the junction at a particular time that can smoothly move

through without any unnecessary delay or congestion. Hence even when the traffic density will be high, cars will keep moving through the junction without causing congestion. The data collected at every light will be processed in real time. The road with the maximum number of vehicles in waiting will be given a higher priority and all the other traffic light timings will be adjusted accordingly. The system will automatically adjust and adapt based on the traffic density on each road at different times of the day like during rush hours in morning and evening or during festive season when the traffic is more than usual. This will ensure an overall smooth and the most efficient traffic movement possible through the junction at any particular time.

IV. Benefits of implementing this system

Traffic jam is a very serious problem which contributes to other major issues we are facing today like pollution, wastage of fuel and energy etcetera. Since most of the people have to deal with it on a daily basis they get psychologically affected and it creates a negative impact on the mind of a person. It also negatively affects work, education and has degrading effect on personal lives of people and finally to the progress of our country. Traffic jams results in non-productive activity and time wastage. People experience delays for their important work. This results in personal as well as professional losses. The traffic congestion and constant blowing of horns create excessive noise pollution. It increases stress and frustration among motorists and passengers which leads to unsafe driving, road mishaps and injuries. Businesses nowadays provide home delivery services. Such time bound businesses are strongly affected by the traffic jam. Emergency vehicles like fire brigade and ambulance get stuck up in traffic jams that cause a delay in reaching the location thereby endangering lives. With the implementation of our proposed system, traffic jams at junctions can be considerably decreased and in many cases can be totally avoided which will prevent all the above mentioned problems that are a by product of these traffic jams and congestions.

V. Conclusion

Traffic jam is a serious issue in every big city of India that causes several problems for common people. It is important for the government to execute proper planning and come up with new and effective strategies like ours which will help in reducing the issue of traffic congestion. Our idea can help in enhancing the traffic management system which will make the execution of traffic related tasks much easier resulting in a smooth and efficient flow of traffic through busy junctions. Moreover, the government needs to utilize this latest and innovative technique in order to reduce the adverse effects of

traffic jam on the roads. It consumes so much of time and energy unnecessarily and hence the loss of the nation.

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